

Counting Automorphic Schur Rings

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Definition of a Group

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Definition 1.4 Isomorphic Two groups are isomorphic if they are the same up to renaming things. Denoted by: $(\cong)$

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Theorem 1.3

$\mathbb{Z}_n = \langle a \rangle$ has a unique subgroup for each divisor k of n .

Definition of Schur Rings

Definition 1.6 (Schur Ring). Let $C_1, C_2, \dots, C_r$ be a partition of a finite group G and let S be the subspace of $F[G]$ spanned by $\overline{C_1}, \overline{C_2}, \dots, \overline{C_r}$. We say that S is a **Schur Ring** over G if

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- $C_1 = 1$;
- for each i there is a j such that $C_i^* = C_j$;
- for each i and j we have $\overline{C_i} \cdot \overline{C_j} = \sum_{k=1}^r \lambda_{i,j,k} \overline{C_k}$, for constants $\lambda_{i,j,k} \in F$.

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- Automorphic

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Thm 1.4

For a finite cyclic group, $\mathbb{Z}_n$ there is a 1-1 correspondence between the subgroups, H , of $\text{Aut}(\mathbb{Z}_n)$ and Automorphic Schur rings of $\mathbb{Z}_n$.

Illustrative example

A good example is $\mathbb{Z}_{15}$. Since

$$\mathrm{Aut}(\mathbb{Z}_{15}) \cong \mathrm{Aut}(\mathbb{Z}_5) \times \mathrm{Aut}(\mathbb{Z}_3) \cong \mathbb{Z}_4 \times \mathbb{Z}_2$$

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We count the subgroups of $\text{Aut}(\mathbb{Z}_{15})$ using the lattice of subgroups of $\mathbb{Z}_4 \times \mathbb{Z}_2$.

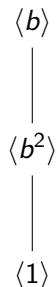
Lattice of $\text{Aut}(\mathbb{Z}_{15})$

Let a be a generator of $\mathbb{Z}_2$, and let b be the generator of $\mathbb{Z}_4$.

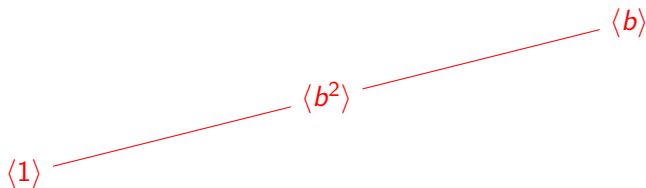
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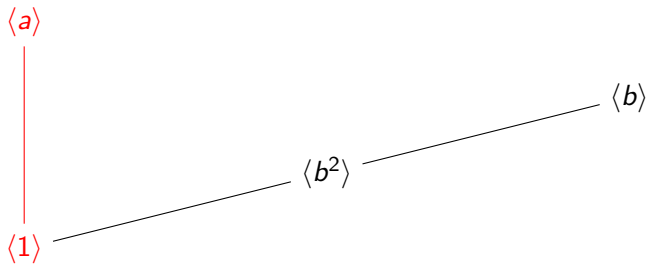
Lattice of $\mathbb{Z}_2$, $L(\mathbb{Z}_2)$ Lattice of $\mathbb{Z}_4$, $L(\mathbb{Z}_4)$



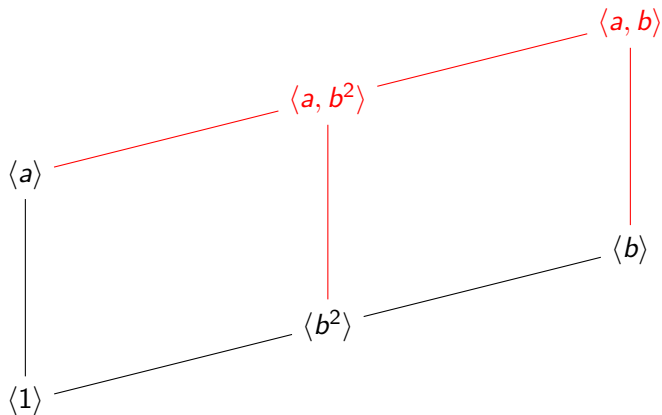
$$L(\mathbb{Z}_2) \times L(\mathbb{Z}_4)$$



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Diagonal Subgroups

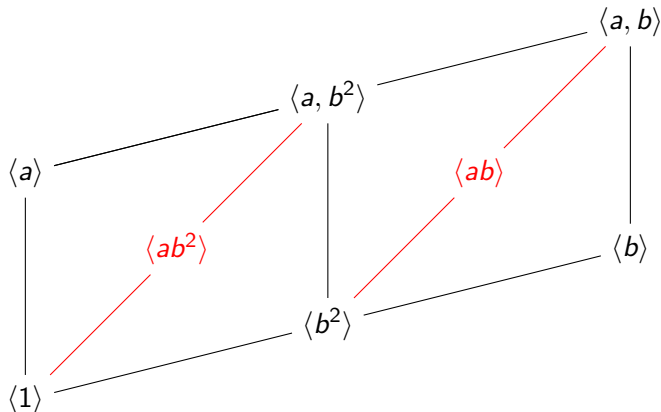
Definition 1.7 (Diagonal Subgroup). For a group $G = H \times K$, and $D \leq G$, D forms a **Diagonal Subgroup** if $D \in \{L(G) \setminus [L(H) \times L(K)]\}$

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These are easily found because there is a bijective correspondence between the diagonal subgroups and the isomorphisms between quotient subgroups of H and quotient subgroups of K .

With this definition we can finish our picture of $L(\mathbb{Z}_4 \times \mathbb{Z}_2)$:



Generalized Results

Number of Automorphic Schur Rings of $\text{Aut}(\mathbb{Z}_{p^k})$. (A. Misseldine)

The number of Automorphic Schur Rings is nk , where n is the number of divisors of $p - 1$.

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By Thm 1.1 we only need to consider $\mathbb{Z}_{(p-1)p^{k-1}}$ and its subgroups. The number of subgroups is equivalent to factors of $(p - 1)p^{k-1}$. p^{k-1} has k factors. Therefore, $\text{Aut}(\mathbb{Z}_{p^k})$ has nk subgroups, so there are nk Automorphic Schur rings.

Generalized Results

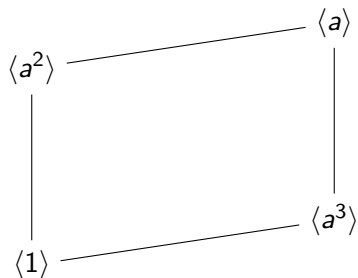
Number of Automorphic Schur Rings of $\text{Aut}(\mathbb{Z}_{2p^k})$. (K.M.S)

The number of Automorphic Schur Rings is nk , where n is the number of divisors of $p - 1$.

$$\text{Aut}(\mathbb{Z}_{2p^k}) \cong \text{Aut}(\mathbb{Z}_2 \times \mathbb{Z}_{p^k}) \cong \text{Aut}(\mathbb{Z}_2) \times \text{Aut}(\mathbb{Z}_{p^k}) \cong \text{Aut}(\mathbb{Z}_{p^k})$$

A quick example

Consider $\mathbb{Z}_{18}$. $\text{Aut}(\mathbb{Z}_{18}) \cong \mathbb{Z}_6$. Let a be the generator.



In the case of the subgroups of $\mathbb{Z}_{pq}$:

$$\text{Aut}(\mathbb{Z}_{pq}) \cong \text{Aut}(\mathbb{Z}_p) \times \text{Aut}(\mathbb{Z}_q) \cong \mathbb{Z}_{p-1} \times \mathbb{Z}_{q-1}$$

Specific prime cases

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A prime p is a **Fermat Prime** if $p = 2^k + 1$.

A prime p is a **Safe Prime** if $p = 2q + 1$ where q is a prime.

$\text{Aut}(\mathbb{Z}_{pq})$, (p, q both Fermat Primes)

Automorphic S-Rings for Fermat Primes. (K.M.S)

For $p = 2^k + 1, q = 2^l + 1$ both Fermat primes, $k \neq l$

Assuming $k < l$, the number of automorphic S-Rings of $\mathbb{Z}_{pq}$ is

$$(k+1)(l+1) + \sum_{n=0}^{k-1} 2^n (k-n)(l-n)$$

Fermat Table

pq	3	5	17	257	65537
3	—	8	14	26	50
5		—	29	57	113
17			—	207	455
257				—	5603
65537					—

$\text{Aut}(\mathbb{Z}_{pq})$, (p, q both Safe Primes)

Automorphic S-rings of $\mathbb{Z}_{pq}$, p, q both Safe Primes. (K.M.S)

For $p = 2r + 1$, $q = 2s + 1$, where r, s are primes, $r \neq s$
 $\text{Aut}(\mathbb{Z}_{pq}) \cong \mathbb{Z}_{2r} \times \mathbb{Z}_{2s}$. The number of Automorphic S-rings of
 $\mathbb{Z}_{2r} \times \mathbb{Z}_{2s}$ is

$$|\text{L}(\mathbb{Z}_{2r}) \times \text{L}(\mathbb{Z}_{2s})| + 1$$

$\text{Aut}(\mathbb{Z}_{pq})$, One Safe Prime, One Fermat Prime

Automorphic S-rings of $\mathbb{Z}_{pq}$, One Safe Prime, One Fermat Prime. (K.M.S)

Let $p = 2^k + 1$, $q = 2r + 1$, where r is prime.

$\text{Aut}(\mathbb{Z}_{pq}) \cong \mathbb{Z}_{2^k} \times \mathbb{Z}_{2r}$ The number of Automorphic S-rings of $\mathbb{Z}_{2^k} \times \mathbb{Z}_{2r}$ is

$$(k+1)(4) + k = 5k + 4$$

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Thank you for listening.